KUTAISI INTERNATIONAL UNIVERSITY

Numerical Differential Equations with AI

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#1.3

Existence and Uniqueness of ODE solutions

- Peano Theorem
- Picard-Lindeloeff Theorem
- Osgood uniqueness criterion
- Q & A

Peano Existence Theorem

Theorem 1.1 (Peano Existence Theorem)

Consider the initial value problem (IVP):

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0$$

where $f : D \subseteq \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ *is continuous on a domain* D.

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- **Key Idea:** Continuity of f is sufficient for the existence of a solution.
- **Note:** Does not guarantee uniqueness.

Arzelà-Ascoli Theorem

Theorem 1.2 (Arzelà-Ascoli Theorem)

Let F be a family of real-valued continuous functions defined on a closed and bounded interval $I \subseteq \mathbb{R}$.
If F is:

- ***Equicontinuous**: For every $\epsilon > 0$, there exists $\delta > 0$ such that for all $f \in F$ and all $x, y \in I$ with $|x - y| < \delta$, we have $|f(x) - f(y)| < \epsilon$.*

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Then, F contains a uniformly convergent subsequence.

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- **Significance:** Essential in the proof of Peano's existence theorem.

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Then, F contains a uniformly convergent subsequence.

- **Key Idea:** Provides conditions for the existence of a uniformly convergent subsequence of continuous functions.
- **Significance:** Essential in the proof of Peano's existence theorem.
- **Application in Peano's Theorem:** The sequence of approximate solutions $\{y_n(t)\}$ is shown to be equicontinuous and uniformly bounded, allowing us to extract a convergent subsequence.

Proof of Peano Theorem (Part 1)

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- Construct a sequence of approximate solutions using Euler's method:

$$y_n(t) = y_0 + \int_{t_0}^{t} f(s, y_n(s_n))ds$$

where s_n is a piecewise constant approximation of $y_n(s)$.

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- Show that the sequence $\{y_n(t)\}$ is uniformly bounded and equicontinuous on $[t_0 - \alpha, t_0 + \alpha]$.

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- This completes the proof.

Picard-Lindelöf Theorem

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If f *is Lipschitz continuous with respect to* y *on* D, *then there exists a unique solution* $y(t)$ *to the IVP on some interval* $[t_0 - \alpha, t_0 + \alpha]$, *where* $\alpha > 0$.

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- **Key Idea:** Lipschitz continuity of f guarantees both existence and uniqueness.

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- **Method of Proof:** Uses the method of successive approximations (Picard iteration).
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- **Limitation:** Requires stronger condition (Lipschitz continuity).

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- Define the Picard iteration sequence:

$$y_0(t) = y_0$$

$$y_{n+1}(t) = y_0 + \int_{t_0}^{t} f(s, y_n(s))ds, \quad n = 0, 1, 2, \ldots$$

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- Choose $\alpha = \min\{a, b/M\}$, where $M = \max_{(t,y)\in R} |f(t, y)|$. This ensures $y_n(t)$ stays in R.

Picard-Lindelöf Theorem Proof (Part 2)

- Show that the sequence $\{y_n(t)\}$ is a Cauchy sequence in the space of continuous functions on $[t_0 - \alpha, t_0 + \alpha]$ with the uniform norm.

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- Estimate the difference between successive iterations:

$$\begin{aligned} |y_{n+1}(t) - y_n(t)| &\leq \int_{t_0}^{t} |f(s, y_n(s)) - f(s, y_{n-1}(s))| ds \\ &\leq L \int_{t_0}^{t} |y_n(s) - y_{n-1}(s)| ds \end{aligned}$$

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- By induction, show that:

$$|y_{n+1}(t) - y_n(t)| \leq \frac{ML^n |t - t_0|^{n+1}}{(n+1)!} \leq \frac{ML^n \alpha^{n+1}}{(n+1)!}$$

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- Since $\sum_{n=0}^{\infty} \frac{L^n \alpha^{n+1}}{(n+1)!}$ converges (it's related to $e^{L\alpha}$), the sequence $\{y_n(t)\}$ is Cauchy.

Picard-Lindelöf Theorem Proof (Part 3)

- Since the space of continuous functions is complete, the Cauchy sequence $\{y_n(t)\}$ converges uniformly to a continuous function $y(t)$.

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- Differentiating with respect to t, we get $\frac{dy}{dt} = f(t, y(t))$ with $y(t_0) = y_0$, so $y(t)$ is a solution.
- Uniqueness follows from the Lipschitz condition. If $y_1(t)$ and $y_2(t)$ are two solutions, then:

$$|y_1(t) - y_2(t)| \leq L \int_{t_0}^{t} |y_1(s) - y_2(s)|ds$$

By Gronwall's inequality, $y_1(t) = y_2(t)$.

Osgood Uniqueness Criterion

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Consider the IVP:

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If there exists a continuous, non-decreasing function $\omega : [0, \infty) \to [0, \infty)$ *such that* $\omega(0) = 0$, $\omega(u) > 0$ *for* $u > 0$, $\int_0^\epsilon \frac{du}{\omega(u)} = \infty$ *for some* $\epsilon > 0$, *and*

$$\|f(t, y_1) - f(t, y_2)\| \leq \omega(\|y_1 - y_2\|)$$

for all $(t, y_1), (t, y_2) \in D$, *then the solution to the IVP is unique.*

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- **Key Idea:** Provides a weaker uniqueness condition than Lipschitz continuity.
- **Significance:** Allows for uniqueness in cases where f is not Lipschitz continuous, but is "almost" Lipschitz.
- **Example:** $\omega(u) = u \log(1/u)$ satisfies the Osgood condition but is not Lipschitz near $u = 0$.

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where ω is continuous, non-decreasing, $\omega(0) = 0$, $\omega(u) > 0$ for $u > 0$, and $\int_0^\epsilon \frac{du}{\omega(u)} = \infty$.

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- Suppose $y_1(t)$ and $y_2(t)$ are two solutions to the IVP.
- Let $u(t) = \|y_1(t) - y_2(t)\|$. Then $u(t_0) = 0$ and $u(t) \geq 0$.

Osgood Uniqueness Criterion Proof (Part 1)

- Consider the IVP: $\frac{dy}{dt} = f(t, y)$, $y(t_0) = y_0$.
- Assume f is continuous and satisfies Osgood's condition:

$$\|f(t, y_1) - f(t, y_2)\| \leq \omega(\|y_1 - y_2\|)$$

  where ω is continuous, non-decreasing, $\omega(0) = 0$, $\omega(u) > 0$ for $u > 0$, and $\int_0^\epsilon \frac{du}{\omega(u)} = \infty$.
- Suppose $y_1(t)$ and $y_2(t)$ are two solutions to the IVP.
- Let $u(t) = \|y_1(t) - y_2(t)\|$. Then $u(t_0) = 0$ and $u(t) \geq 0$.
- We have:

$$\frac{du}{dt} = \frac{d}{dt}\|y_1(t) - y_2(t)\| \leq \|f(t, y_1(t)) - f(t, y_2(t))\| \leq \omega(u(t))$$

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- However, $\int_0^{\epsilon} \frac{du}{\omega(u)} = \infty$ for some $\epsilon > 0$. Therefore, if $u(t) > 0$ for some t, the left-hand side would become infinite, contradicting the inequality.
- Thus, $u(t) = 0$ for all t, which implies $y_1(t) = y_2(t)$. This shows uniqueness.

Order Reduction: Higher to First Order (Part 1)

- Consider an n-th order ordinary differential equation:

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- This results in a system of n first-order ODEs:

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- In vector notation, let $\mathbf{y} = (y_1, y_2, \ldots, y_n)^T$. Then:

$$\mathbf{y}' = \mathbf{F}(t, \mathbf{y})$$

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- Initial conditions for the n-th order ODE, $y(t_0) = c_1, y'(t_0) = c_2, \ldots, y^{(n-1)}(t_0) = c_n$, become initial conditions for the first-order system: $\mathbf{y}(t_0) = (c_1, c_2, \ldots, c_n)^T$.

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- This transformation allows us to apply existence and uniqueness theorems (like Picard-Lindelöf or Peano) to higher-order ODEs by analyzing the equivalent first-order system.

Q & A